VOL. I · NO. 1 MONDAYS & THURSDAYS

THE LAUNCH REPORT

Rocket launches, reentries and the business of spaceflight, worldwide

THURSDAY, OCTOBER 8, 2026 ALL TIMES EASTERN DAYLIGHT TIME (EDT)

THE WEEK IN LAUNCH

Nuri goes four for four, Crew-12 comes home, and a Mars-moon mission heads for the pad

South Korea's Nuri rocket put 14 satellites in orbit on Wednesday, its fourth success in a row. It was the only orbital launch anywhere in the world between Monday and Thursday morning, according to Launch Library 2. A SpaceX Falcon 9 carrying 21 Space Development Agency (SDA) satellites aborted at T-minus one minute on Monday and is now set for 3:29 a.m. EDT Saturday. On Thursday, the Crew-12 astronauts splashed down off Los Angeles after 237 days on the International Space Station (ISS).

The next two weeks are busier: a Long March 12 from Hainan on Friday, three more Falcon 9 launches, and NASA's 35th Commercial Resupply Services cargo flight (CRS-35) on Tuesday. Then, on October 19, Japan's H3 rocket launches the Martian Moons eXploration (MMX) probe, which is designed to return samples from the Martian moon Phobos. Three old rocket stages are predicted to fall back to Earth on Saturday. In Washington, the Federal Aviation Administration (FAA) has proposed loosening five launch-safety rules.

I · RECENT LAUNCHES & RETURNS

Nuri goes four for four

South Korea's homegrown rocket delivered five research microsatellites and nine CubeSats; one CubeSat did not separate.

Nuri flight 5: success from Naro

The three-stage Nuri, also called the Korea Space Launch Vehicle-II (KSLV-II), lifted off on schedule from Naro Space Center at 12:25 p.m. Korea Standard Time (KST) on Wednesday, October 7, which was 11:25 p.m. EDT Tuesday (03:25 UTC). It carried five NEONSAT Earth-observation microsatellites, numbers 2 to 6, built by the Korea Advanced Institute of Science and Technology (KAIST), each under 100 kg, plus 10 CubeSats, bound for a sun-synchronous orbit about 575 km up, Yonhap reported. All five NEONSATs had contacted ground stations by 5:52 p.m. KST, according to the Korea Herald.

Nine of the 10 CubeSats deployed. The tenth, PERSAT02 from the startup Quaternion, failed to separate (Yonhap). The Korea Aerospace Research Institute (KARI) runs the program with Hanwha Aerospace. Nuri now has four straight successes and a record of four out of five, after its first flight failed in 2021 (Space.com). The next flight, carrying NEONSATs 7 to 11, is planned for the second half of 2027. The government has added five more Nuri flights through 2032, budgeted at 328 billion won (Korea Herald).

Falcon 9 aborts at T-minus one minute

SpaceX's first try at the SDA's Tranche 1 Transport Layer A mission from Vandenberg Space Force Base, California, ended on Monday, October 5. The countdown stopped just after T-minus one minute, as the vehicle entered startup. No reason was given, Spaceflight Now reported. A second attempt on October 6 was waved off, and the launch then moved to October 8 and October 9. In a post on X on Wednesday evening, SpaceX said it was now targeting October 10:

> *"The rocket and payload remain healthy, and additional time allows for double checks on vehicle systems and range deconfliction"*
>
> SPACEX · @SPACEX, WED OCT 7, 8:34 P.M. EDT

Crew-12 splashes down after 237 days

SpaceX's Crew Dragon Freedom undocked from the space-facing port of the station's Harmony module at 8:05 a.m. EDT Wednesday, NASASpaceflight reported. It splashed down in the Pacific off Los Angeles at 11:34 a.m. EDT Thursday and was recovered by the ship Shannon, NASA said. Aboard were NASA astronauts Jessica Meir and Jack Hathaway, European Space Agency (ESA) astronaut Sophie Adenot and Roscosmos

cosmonaut Andrey Fedyaev. The mission lasted 237 days, and Meir now has 440 days in space across her flights. It was Freedom's fifth mission. SpaceX posted, "Splashdown of Dragon confirmed!"

II · LAUNCH SCHEDULE

Seven launches in 11 days, from Hainan to Tanegashima

Every orbital launch with a date between now and October 22, plus missions targeting later this month. Times are EDT, with Coordinated Universal Time (UTC) below; NET means no earlier than.

DATE / TIME	VEHICLE	PROVIDER	MISSION / PAYLOAD	SITE	ORBIT	STATUS	SOURCES
Fri Oct 9, 3:25 p.m. 19:25 UTC	Long March 12	China Aerospace Science and Technology Corp. (CASC)	Not announced; NASASpaceflight expects a SatNet (Guowang) batch	Commercial LC-2, Wenchang, China	—	**TENTATIVE**	Launch Library 2 (from airspace notices); NASASpaceflight
Sat Oct 10, 3:29 a.m. 07:29 UTC	Falcon 9	SpaceX	SDA Tranche 1 Transport Layer A: 21 Northrop Grumman satellites. Booster B1103, 6th flight	SLC-4E, Vandenberg, Calif.	Polar LEO	**CONFIRMED**	SpaceX; FAA advisory; Spaceflight Now
Mon Oct 12, 7:00 p.m. 23:00 UTC	Falcon 9	SpaceX	Starlink 15-25: 27 Starlink V2 Mini satellites. Booster B1088, 20th flight; drone-ship landing	SLC-4E, Vandenberg, Calif.	LEO, 70°	**TENTATIVE**	Spaceflight Now; Launch Library 2
Tue Oct 13, 6:33 a.m. 10:33 UTC	Falcon 9 / Cargo Dragon	SpaceX for NASA	CRS-35: more than 6,300 lb of cargo for the ISS. Booster B1096, 8th flight; landing back at the Cape	SLC-40, Cape Canaveral, Fla.	LEO (ISS)	**CONFIRMED**	NASA; FAA advisory
Thu Oct 15, 7:44 p.m. 23:44 UTC	Falcon 9	SpaceX for the U.S. Space Force	USSF-481. Booster B1081, 28th flight	SLC-4E, Vandenberg, Calif.	Polar LEO	**TENTATIVE**	Launch Library 2 (from maritime warnings)
Oct 15–20 target	GSLV Mk II	Indian Space Research Organisation (ISRO)	NVS-03 (IRNSS-1L) navigation satellite	Second Launch Pad, Sriharikota, India	GTO	**PRELIMINARY**	Times of India; Launch Library 2
Mon Oct 19, 3:41:03 p.m. 19:41:03 UTC	H3-24	Mitsubishi Heavy Industries (MHI) for JAXA	Martian Moons eXploration (MMX), Phobos sample-return probe	LP-2, Yoshinobu complex, Tanegashima, Japan	Mars transfer	**CONFIRMED**	JAXA; launch plan
NET Oct 29	Vulcan VC6L	United Launch Alliance (ULA)	Amazon Leo LV-01 broadband satellites	SLC-41, Cape Canaveral, Fla.	LEO	**PRELIMINARY**	Launch Library 2; Ars Technica
NET October	Ariane 64	Arianespace	Amazon Leo LE-04	ELA-4, Kourou, French Guiana	LEO	**PRELIMINARY**	Launch Library 2
NET October	Falcon 9	SpaceX	Bandwagon-5 rideshare	SLC-40, Cape Canaveral, Fla.	Mid-inclination LEO	**PRELIMINARY**	Launch Library 2
NET October	Long March 12A	CASC	Flight 2	Jiuquan, China	LEO	**PRELIMINARY**	Launch Library 2
NET October	SSLV	ISRO	AayulSAT-1A and others	First Launch Pad, Sriharikota, India	LEO	**PRELIMINARY**	Launch Library 2

CONFIRMED date and time announced by the launch provider, its customer or the FAA. **TENTATIVE** specific date in published schedules or airspace and maritime notices, not yet confirmed by the operator. **PRELIMINARY** target range, or no-earlier-than date or month only.

Abbreviations: LC, launch complex; SLC, Space Launch Complex; SLC-4E, Space Launch Complex 4 East; LP-2, Launch Pad 2; ELA-4, Ariane Launch Area 4; LEO, low Earth orbit; GTO, geostationary transfer orbit; GSLV, Geosynchronous Satellite Launch Vehicle; SSLV, Small Satellite Launch Vehicle; JAXA, Japan Aerospace Exploration Agency.

Date changes

SDA Tranche 1 Transport Layer A. First set for Mon Oct 5 (abort at T-1 minute), then Oct 6, 8 and 9; now Sat Oct 10. SpaceX cited double checks of vehicle systems and range deconfliction (SpaceX; Spaceflight Now).

Starlink 15-25. First listed for Sep 30, then Oct 10 and Oct 11; now Mon Oct 12. No reason given (Spaceflight Now; Launch Library 2).

What's flying

Long March 12. The launch was added to schedules after airspace notices appeared. If it flies, it would be the fifth Long March 12 of the year, on a southeasterly track out of Wenchang (NASASpaceflight). The window runs from 3:20 to 3:49 p.m. EDT.

QUICK HIT

Starship. NASASpaceflight's unofficial schedule puts Flight 15 from Starbase Pad 2 NET November 8 and Flight 16, possibly from Kennedy's Launch Complex 39A, NET December 10.

SDA Transport Layer. This is the fourth of six Tranche 1 Transport Layer launches. The satellites are part of the SDA's data transport layer, and 63 Tranche 1 satellites are already up, according to Spaceflight Now. The launch time is instantaneous.

CRS-35. The 35th SpaceX cargo flight to the ISS uses Cargo Dragon C211 on its fourth flight. It carries the final pair of ISS Roll-Out Solar Arrays, along with hardware for making artificial retinas, 3D heart-tissue models, glass-formation samples and brain organoids for research on neurodegenerative diseases (NASA).

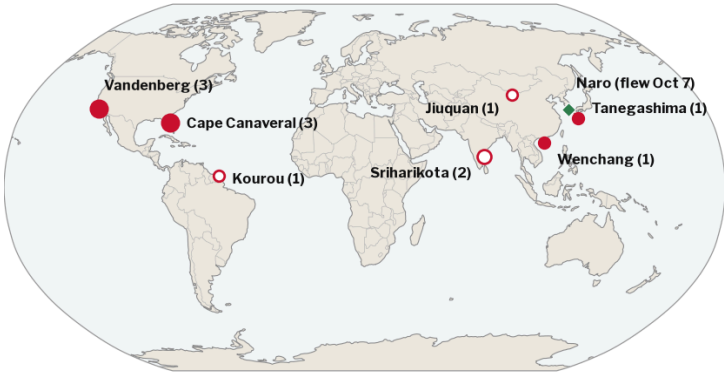
H3 / MMX. JAXA's probe is designed to bring Phobos samples back to Earth in Japanese fiscal year 2031, which would be the first sample return from the Mars system (JAXA launch plan). If the launch slips, the reserve period runs to November 7, with a new launch time set for each day. Liftoff is 4:41:03 a.m. Japan Standard Time (JST) on Tuesday, October 20.

NVS-03. India's NavIC navigation system has only three working satellites, one short of the four it needs for standalone positioning, Union minister Jitendra Singh said. ISRO's previous navigation satellite, NVS-02, was stranded after its 2025 launch because a signal failed to reach a pyro valve in its orbit-raising engine (Times of India).

Vulcan returns. "I can say we're weeks away from launching on Vulcan," Amazon Leo chief Rajeev Badyal told Ars Technica. He said an Arianespace launch would follow, with at least one more Vulcan this year, and that Vulcan's nozzle issue "has been fixed."

Further out

China's Shenzhou-24 crew launch looks likely around November, according to SpaceNews.



Launch sites with orbital launches scheduled through October 31; numbers count launches. Filled: dated launch (Confirmed or Tentative). Hollow: Preliminary. Green diamond: Nuri's October 7 launch site.

MAP: THE LAUNCH REPORT · PAD COORDINATES FROM LAUNCH LIBRARY 2 · NATURAL EARTH

III · TRAJECTORIES & HAZARD AREAS

Where H3's spent stages will fall

JAXA has published the sea areas where MMX's boosters, nose fairing and first stage are expected to come down. FAA airspace windows for this week's U.S. launches are below.

H3 Flight 10 timeline

According to the launch plan, the rocket turns its pitch plane to an azimuth of 90 degrees, due east, shortly after liftoff and flies out over the Pacific. The upper stage fires twice and releases MMX into a Mars transfer orbit 71 minutes after liftoff. The times below assume launch on October 20 JST. Because the probe is headed to Mars, events after the first second-stage cutoff shift with the launch date.

EVENT	TIME AFTER LIFTOFF	ALTITUDE
Solid rocket boosters separate (two pairs)	1:56 / 1:58	59–61 km
Payload fairing separates	3:17	131 km
Main engine cutoff (MECO)	4:52	208 km
First and second stages separate	4:59	214 km
Second-stage engine start 1 (SELI1)	5:12	224 km
Second-stage engine cutoff 1 (SECO1)	10:40	329 km
Second-stage engine start 2 (SELI2)	1:03:57	322 km
Second-stage engine cutoff 2 (SECO2)	1:10:21	497 km
MMX separates, 11.2 km/s	**1:11:12**	**592 km**

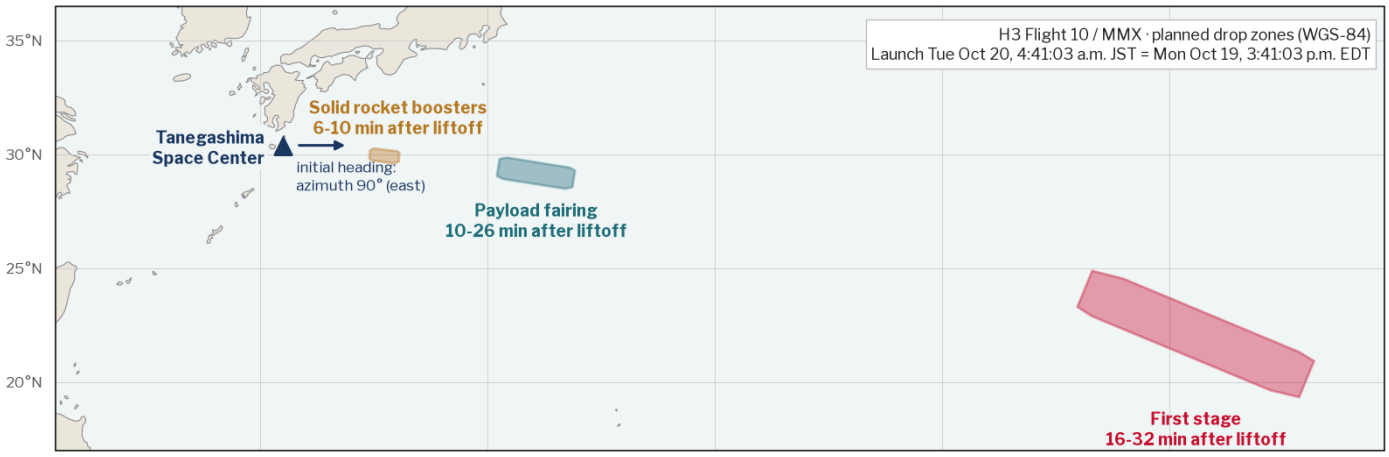


JAXA's HTV-X1 cargo spacecraft approaches the International Space Station on Oct. 29, 2025, four days after launching from Tanegashima on an H3-24, the same H3 version that will carry MMX.

PHOTO: NASA · PUBLIC DOMAIN

Flight paths

From Vandenberg, the SDA mission will fly a southerly track into polar orbit, and Starlink 15-25 will head southeast toward a 70-degree orbit. The Long March 12 is also expected to fly southeast from Wenchang (NASASpaceflight).



Planned drop zones for H3 Flight 10, plotted from the corner coordinates in JAXA's launch plan (World Geodetic System 1984). The solid rocket booster zone measures 62 by 128 km, the fairing zone 107 by 330 km and the first-stage zone 226 by 1,105 km. Ships and aircraft are warned to stay out of these zones through Japan's coast guard and transport ministry. The arrow shows the eastward heading the plan gives for shortly after liftoff; it is not a ground track.

MAP: THE LAUNCH REPORT · DATA FROM THE JAXA H3/MMX LAUNCH PLAN (AUG. 20, 2026) · NATURAL EARTH

U.S. airspace windows

The FAA Air Traffic Control System Command Center's operations plan advisory 052, issued at 1:25 p.m. EDT Thursday, lists these launch windows:

LAUNCH	PRIMARY (EDT)	BACKUP (EDT)
Northrop Grumman MK21-2B, Vandenberg	Fri Oct 9, 2:01–8:34 a.m.	Sat Oct 10, 2:01–8:34 a.m.
SpaceX SDA-T1A, Vandenberg	Sat Oct 10, 3:29–4:34 a.m.	Sun Oct 11, 3:21–4:26 a.m.
SpaceX CRS-35, Cape Canaveral	Tue Oct 13, 6:24–7:14 a.m.	Wed Oct 14, 5:58–6:48 a.m.

The same advisory still listed Starlink 15-25 for Saturday evening, October 10. Spaceflight Now and Launch Library 2 have since moved it to Monday.

IV · REENTRIES & ISS TRAFFIC

Three old rocket stages are due down Saturday

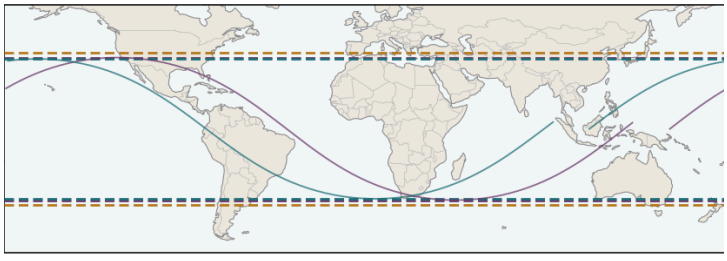
A Delta 2 stage from 1990 is among the objects predicted to reenter. At the station, one Dragon has just left and the next arrives on October 15.

Predicted reentries

The Aerospace Corporation's reentry database predicts these three rocket bodies will fall on Saturday, October 10. Uncertainties of 14 to 16 hours mean each object could come down anywhere along many orbits.

OBJECT	LAUNCHED	PREDICTED (EDT)
Electron upper stage, "Owl Around The World"	Sep 2, 2026	Sat 2:27 a.m. ± 15 hr
Delta 2 first stage, NAVSTAR 18 GPS mission	Jan 24, 1990	Sat 11:43 a.m. ± 14 hr
Pegasus upper stage, CYGNSS weather satellites	Dec 15, 2016	Sat 4:51 p.m. ± 16 hr

The latest orbits published by CelesTrak are 220 by 231 km for the Delta 2 stage, 212 by 224 km for the Pegasus stage and 186 by 315 km for the Electron stage.



Each object can come down only between the dashed lines of its color, set by its orbital inclination. The solid curves show one orbit of the Delta 2 and Pegasus stages, computed from orbital elements of October 8; the reentry point itself cannot be predicted this far ahead.

MAP: THE LAUNCH REPORT · DATA FROM CELESTRAK AND AEROSPACE CORP. · NATURAL EARTH

Starlink satellites coming down

The same database lists 16 Starlink satellites reentering between October 5 and 8, seven of them confirmed down. CelesTrak's catalog records Starlink-1272, launched in March 2020, and Starlink-1448 as decayed on October 6.



A SpaceX cargo Dragon backs away from the International Space Station after undocking from the Harmony module's forward port on June 16, 2026, above a cloudy North Pacific.

PHOTO: NASA · PUBLIC DOMAIN

Station traffic

Crew-12's departure on Wednesday cleared Harmony's space-facing port, where CRS-35 will dock on its own at about 11 a.m. EDT on Thursday, October 15. The cargo Dragon will stay until mid-November and then return to a splashdown off California with time-critical research (NASA). The next Russian cargo ship, Progress MS-36, is listed for 12:03 a.m. Eastern Standard Time (EST) on November 24 from Baikonur (Launch Library 2).

V · INDUSTRY & REGULATION

FAA proposes five changes to launch licensing

The proposals target Part 450, the licensing rule for U.S. commercial launches and reentries in effect since 2021. Separate environmental reviews could allow many more Starship and New Glenn launches from Florida.

Part 450 proposals

Five notices of proposed rulemaking appeared in the Federal Register on October 5, with comments due November 4. They would simplify how operators show compliance with flight safety analysis (docket FAA-2026-9936); add an exception to lightning-hazard rules (FAA-2026-9940); delete the word "hypothetical" from far-field blast analysis, so it covers existing windows only (FAA-2026-9941); simplify physical containment requirements (FAA-2026-9934); and allow fully digital license applications (FAA-2026-9935).

Transportation Secretary Sean Duffy announced the package, the first output of the Department of Transportation's Space Policy Administration, Coordination, and Execution (SPACE) Task Force, Payload reported. The changes are meant to help reach the White House goal of 1,000 commercial launches and reentries a year by 2030. Acting Deputy FAA Administrator Liam McKenna called them "just the beginning."

Environmental reviews in Florida

Starship at SLC-37. The FAA is preparing a Supplemental Environmental Impact Statement (SEIS) on SpaceX's plan for up to 76 Starship launches and 152 landings a year at Cape Canaveral. Scoping comments are due Wednesday, October 14 (notice, docket FAA-2026-10792).

New Glenn at SLC-36. A draft Environmental Assessment (EA) covers Blue Origin's request to raise New Glenn's cap from 12 to 50 launches a year, with no new construction. A virtual public meeting is set for October 22 from 6 to 8 p.m. EDT, and comments close October 30 (notice, docket FAA-2026-9841).

Amazon Leo nears service

Amazon has built its 1,000th Leo broadband satellite and has 396 in orbit. It plans to start initial service in 2026, the company told Ars Technica in an interview published Thursday.

QUICK HIT

Terran R. The FAA has adopted a Finding of No Significant Impact and Record of Decision for Relativity Space's Terran R program at Cape Canaveral, under docket FAA-2026-12046 (notice).

Europe

Themis. ArianeGroup and SSC Space completed a second wet dress rehearsal of the Themis reusable-rocket demonstrator at Esrange, Sweden, on September 29. The vehicle was filled with liquid nitrogen instead of propellant. ESA announced the test on October 7 (European Spaceflight). The European Union has extended its €42.8 million SALTO (reuSable strAtegic space Launcher Technologies & Operations) project, which funds Themis's first two hop tests, to December 31, 2026 (European Spaceflight).

HyImpulse. The German startup received a launch operator license from the UK Civil Aviation Authority (CAA) on October 1. The license covers its SR75 suborbital rocket from SaxaVord, Scotland, but the flight has slipped to spring 2027. The company cited the remaining licensing work and autumn weather (European Spaceflight).

India

Agnikul Cosmos fired its single-piece, 3D-printed Agnite semi-cryogenic engine. The test included thrust sweeps and a run "well beyond the max. burn time of some of our ascent phase engines," as the company prepares for its Agnibaan Mission-02 (India Today).

QUICK HIT

Blue Origin in Texas. Blue Origin announced Constellation Park in Hutto, Texas, a 1.3 million-square-foot site with 2,000 jobs, for its TeraWave and Quartz programs.

237 orbital launch attempts so far in 2026

SpaceX accounts for almost half. China needs at least 31 more launches to pass 100 in a year for the first time.

237	230	115	70	31
orbital launch attempts worldwide, Jan 1–Oct 8	reached orbit	by SpaceX, all successful	by China, four failures	Chinese launches needed in Q4 to pass 100

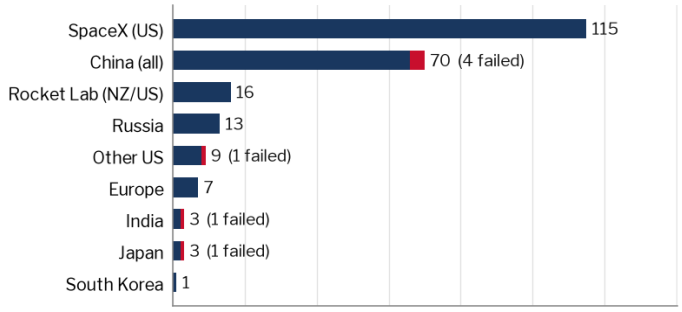
The count

Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) records 236 orbital launch attempts through October 2, 229 of them successful. Nuri's October 7 flight brings the year to 237. The U.S. leads with 125 attempts, followed by China with 70, Rocket Lab's Electron from New Zealand with 15 (plus one from the U.S.) and Russia with 13.

Busiest launch sites

Vandenberg Space Force Base in California is the year's busiest launch site, ahead of Florida's Cape Canaveral and Kennedy Space Center combined. March and April were the busiest months, with 30 orbital launch attempts each (GCAT, through October 2).

SITE	LAUNCHES, 2026
Vandenberg Space Force Base, California	**63**
Cape Canaveral and Kennedy Space Center, Florida	56
Jiuquan, China	26
Mahia, New Zealand (Rocket Lab)	15
Hainan commercial launch site, Wenchang, China	13
Taiyuan, China	12
Plesetsk, Russia	8



Orbital launch attempts in 2026 by provider or country. Navy: success. Red: failure or partial failure.

CHART: THE LAUNCH REPORT · GCAT (JONATHAN MCDOWELL) THROUGH OCT 2, PLUS LAUNCH LIBRARY 2

The failures

According to GCAT, seven launches have failed this year: India's Polar Satellite Launch Vehicle (PSLV) on January 12; China's Long March 3B on January 16, Ceres-2 on January 17, Tianlong-3 on April 3 and Long March 7A on August 10; Japan's Kairos on March 5; and Blue Origin's New Glenn on April 19, recorded as a partial failure. A New Glenn that exploded on its pad on May 29 is not counted as a launch attempt.

China's fourth-quarter push

China reached 70 orbital launch attempts on October 1. To pass 100 in a year for the first time, it needs at least 31 more in the fourth quarter, after 18, 26 and 26 in the first three, SpaceNews reported. Possible first flights include Orienspace's Gravity-2, iSpace's Hyperbola-3, which would attempt first-stage recovery, and Zenk Space's Zhihang-1.

KEY DATES AHEAD (EDT)

Fri Oct 9	Long March 12, Wenchang, 3:25 p.m.
Sat Oct 10	Falcon 9 SDA launch, 3:29 a.m.; three stage reentries predicted
Mon Oct 12	Falcon 9 Starlink 15-25, 7:00 p.m.
Tue Oct 13	CRS-35 launch, 6:33 a.m.
Wed Oct 14	Starship SLC-37 scoping comments due
Thu Oct 15	CRS-35 docks, about 11 a.m.; USSF-481, 7:44 p.m.
Mon Oct 19	H3 / MMX, 3:41 p.m.
Thu Oct 22	New Glenn cadence meeting, 6–8 p.m.

Takeaways

1. Falcon 9's SDA launch: 3:29 a.m. EDT Saturday, after Monday's abort.
2. CRS-35: 6:33 a.m. EDT Tuesday, with the last ISS Roll-Out Solar Arrays; docking October 15.
3. H3 launches Japan's Phobos sample-return probe, MMX, at 3:41 p.m. EDT on October 19.
4. Three rocket stages, one from 1990, are predicted to reenter on Saturday.
5. Comment deadlines: Starship at SLC-37, October 14; New Glenn, October 30; Part 450, November 4.

SOURCES

Schedules: Launch Library 2 (The Space Devs), Spaceflight Now, NASASpaceflight. Agencies: NASA, JAXA, FAA, Federal Register. Tracking: Aerospace Corp. reentry database, CelesTrak, GCAT (Jonathan McDowell). Reporting: Yonhap, Korea Herald, Space.com, SpaceNews, Ars Technica, Payload, European Spaceflight, Times of India, India Today. X posts: @SpaceX, @NASASpaceflight, @blueorigin. Photos: NASA (public domain). Maps: Natural Earth (public domain).

The Launch Report is compiled from official agency releases, regulatory filings, orbital tracking data, public posts on X and published reporting. Quotes are reproduced verbatim unless marked as translated or paraphrased. Launch times change often; check with the launch provider before making plans.